

02-requirements

6. Functional Requirements

This section enumerates the functional requirements (FRs) for Helios, the Autonomous Growth Diagnosis Engine, grouped by capability domain (A–K). Each FR carries a stable ID, requirement statement, rationale, acceptance criteria, and MoSCoW priority (Must / Should / Could / Won't-for-now). Every requirement is written so that an engineer can verify it deterministically against the GA4 dataset `bigquery-public-data.ga4_obfuscated_sample_ecommerce` and the five MCP servers (`warehouse-mcp` , `semantic-mcp` , `stats-mcp` , `experiment-mcp` , `report-mcp`) and seven agents (Orchestrator, Monitor, Decompose, Diagnose, Prescribe, Narrator, Critic).

6.A Data & Semantic Layer

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-A1	The system MUST expose every governed metric (<code>sessions</code> , <code>users</code> , <code>new_users</code> , <code>returning_users</code> , <code>engaged_sessions</code> , <code>engagement_rate</code> , <code>view_item_sessions</code> , <code>add_to_cart_sessions</code> , <code>begin_checkout_sessions</code> , <code>purchasing_sessions</code> , <code>session_conversion_rate</code> , <code>view_to_cart_rate</code> , <code>cart_to_checkout_rate</code> , <code>checkout_to_purchase_rate</code> , <code>cart_abandonment_rate</code> , <code>checkout_abandonment_rate</code> , <code>transactions</code> , <code>revenue</code> , <code>gross_revenue</code> , <code>net_revenue</code> , <code>aov</code> , <code>items_per_transaction</code> , <code>revenue_per_session</code> , <code>revenue_per_user</code>) only through <code>semantic-mcp.get_metric(name)</code> .	Anti-hallucination grounding; the LLM must never author raw metric SQL.	<code>get_metric(name)</code> returns governed SQL for all 24 canonical metrics; unknown names raise <code>MetricNotFound</code> ; no agent code path emits a metric not present in the registry.	Must

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-A2	The system MUST build all SQL via <code>semantic-mcp.build_query(metric, dims, filters, window)</code> , never by string-concatenation in agent code.	Single validated SQL path; verify-then-trust.	100% of executed queries originate from <code>build_query</code> ; a static check fails CI if any agent module contains a raw <code>SELECT</code> against <code>events_*</code> .	Must
FR-A3	The dbt project MUST materialize the canonical models: staging <code>stg_ga4__events</code> , <code>stg_ga4__event_params</code> ; intermediate <code>int_ga4__sessionized</code> , <code>int_ga4__funnel_steps</code> ; facts <code>fct_sessions</code> , <code>fct_funnel</code> , <code>fct_daily_funnel</code> , <code>fct_orders</code> , <code>fct_order_items</code> ; dims <code>dim_users</code> , <code>dim_items</code> , <code>dim_channels</code> , <code>dim_date</code> ; plus a semantic layer under <code>models/semantic</code> .	Reproducible, tested transformation lineage.	<code>dbt build</code> succeeds; all models exist with the exact snake_case names; <code>dbt docs</code> lineage covers source → staging → fact.	Must
FR-A4	<code>stg_ga4__event_params</code> MUST unnest <code>event_params</code> into typed columns and resolve <code>ga_session_id</code> to define the session key (<code>user_pseudo_id</code> , <code>ga_session_id</code>).	Sessionization is the foundation for every funnel metric.	A session row exists for each distinct (<code>user_pseudo_id</code> , <code>ga_session_id</code>); null <code>ga_session_id</code> rows are quarantined and counted.	Must
FR-A5	<code>dim_channels</code> MUST derive GA4 default channel grouping from SESSION-SCOPED <code>event_params</code> source/medium, falling back to user first-touch <code>traffic_source</code> only when session scope is null.	Honors the documented <code>traffic_source</code> gotcha (user-level first-touch, not session).	Channel labels ∈ {Direct, Organic Search, Paid Search, Display, Paid Social, Organic Social, Email, Affiliates, Referral, Other}; a unit test asserts fallback ordering.	Must

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-A6	warehouse- <code>mcp.reconcile(metric, grain)</code> MUST return canonical totals that every fact-derived metric is checked against.	Result reconciliation guardrail.	Every shipped finding's underlying metric matches <code>reconcile</code> within 0.5% tolerance; mismatches block the finding.	Must
FR-A7	The semantic layer MUST publish <code>list_dimensions()</code> returning exactly the canonical dimensions (<code>device_category</code> , <code>operating_system</code> , <code>browser</code> , <code>country</code> , <code>region</code> , <code>channel_group</code> , <code>source</code> , <code>medium</code> , <code>campaign</code> , <code>landing_page</code> , <code>item_category</code> , <code>item_name</code> , <code>is_new_user</code> , <code>day</code> , <code>week</code> , <code>session_number_bucket</code>).	Bounded, governed slicing vocabulary.	<code>list_dimensions()</code> set-equals the canonical list; <code>build_query</code> rejects any other dimension.	Must
FR-A8	The system SHOULD support an incremental dbt strategy keyed on <code>event_date</code> so only new date shards are processed per run.	Cost and runtime control on date-sharded <code>events_YYYYMMDD</code> .	Incremental run scans only shards newer than the last watermark; full-refresh remains available via flag.	Should

6.B Monitoring & Anomaly Detection

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-B1	The Monitor agent MUST scan all canonical metrics over the run window for anomalies using <code>stats-mcp.detect_anomaly(series, method)</code> .	Math is deterministic and lives in code, not the LLM.	Each metric series is tested; the LLM performs no arithmetic; method $\in$ {stl-residual, ewma, robust-zscore}.	Must
FR-B2	Anomaly detection MUST account for weekly seasonality and the dataset	Avoid false positives from	Seasonal decomposition applied before flagging;	Must

ID	Requirement	Rationale	Acceptance Criteria	Priority
	window (~2020-11-01 to 2021-01-31, including Black Friday / holiday peaks).	known seasonality.	holiday spikes do not trigger spurious anomalies in the labeled benchmark.	
FR-B3	The Monitor MUST emit, per anomaly, the metric, dimension slice, t0/t1 window, magnitude, and direction, then hand off to Decompose.	Structured handoff for plan-execute-critique.	Anomaly record validates against schema; downstream agents receive typed objects, not prose.	Must
FR-B4	The system MUST consult the Memory suppression list before flagging, so previously-explained anomalies are not re-raised.	Reduce alert fatigue; learning.	A suppressed (metric, segment) pair is skipped; suppression is logged with the prior diagnosis ID.	Should
FR-B5	The Monitor SHOULD rank candidate anomalies by preliminary dollar magnitude to focus the run under the time budget.	Time-to-diagnosis < 5 min/run.	Top-N anomalies by estimated <code>revenue</code> impact are processed first.	Should

6.C Decomposition (Mix-Shift vs Rate-Change)

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-C1	For any aggregate rate change, the Decompose agent MUST call <code>stats-mcp.decompose_change(metric, dim, t0, t1)</code> to split ΔR into mix, rate, and interaction effects.	The technical centerpiece; resolves Simpson's paradox.	Output contains <code>mix_effect</code> , <code>rate_effect</code> , <code>interaction</code> per segment and in total; components sum to ΔR within floating tolerance 1e-6.	Must
FR-C2	The decomposition MUST implement exactly: $\text{mix} = \sum \Delta w_i \cdot r_i(t_0)$, $\text{rate} = \sum w_i(t_0) \cdot \Delta r_i$, $\text{interaction} = \sum \Delta w_i \cdot \Delta r_i$.	Determinism where it matters.	A golden-data unit test reproduces hand-computed effects; identity $\sum \text{mix} + \sum \text{rate} + \sum \text{interaction} = \Delta R$ holds.	Must

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-C3	Decomposition MUST be runnable across every canonical dimension to locate the dominant driver dimension.	Attribute the change to the right slice.	The dimension maximizing absolute contribution is reported with its segment-level breakdown.	Must
FR-C4	The system SHOULD detect and flag Simpson's-paradox cases where the aggregate moves opposite to most segments.	Headline insight; defends against naive reads.	When $\text{sign}(\Delta R)$ differs from the majority of $\text{sign}(\Delta r_i)$, a paradox flag is set on the finding.	Should

6.D Diagnosis / Root-Cause Analysis

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-D1	The Diagnose agent MUST build a hypothesis tree and verify each branch with governed SQL via <code>semantic-mcp.build_query</code> + <code>warehouse-mcp.run_query</code> .	Grounding over generation.	Every leaf hypothesis maps to ≥ 1 executed governed query; unverified branches are pruned, not reported.	Must
FR-D2	Each hypothesis MUST carry a significance test from <code>stats-mcp.significance_test(a, b)</code> before promotion to a finding.	Statistical defensibility.	Findings include test statistic, p-value, and effect size; non-significant branches are discarded or marked exploratory.	Must
FR-D3	Every finding MUST pass the Critic agent, which actively attempts to REFUTE it (mix-shift confound, insufficient sample, seasonality, data quality).	Adversarial verify-then-trust.	A finding ships only with a recorded Critic verdict = "survived" and the refutation attempts it withstood.	Must
FR-D4	Diagnosis MUST distinguish whether a funnel movement is driven by mix-shift vs rate-change by consuming FR-C1 output, never asserting cause from aggregates alone.	Core thesis.	Each diagnosis cites the decomposition result; aggregate-only causal claims are blocked by the Critic.	Must

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-D5	The system SHOULD chain funnel steps (<code>session_start</code> → <code>view_item</code> → <code>add_to_cart</code> → <code>begin_checkout</code> → <code>add_shipping_info</code> → <code>add_payment_info</code> → <code>purchase</code>) to localize the leakiest step.	Precise root cause.	The step with the largest adverse delta in step-to-step rate is identified and reported.	Should

6.E Quantification (Revenue-at-Risk)

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-E1	Every shipped finding MUST carry a dollar revenue-at-risk computed from governed <code>revenue</code> , <code>aov</code> , and affected session/conversion volumes.	"Every finding carries a dollar impact" principle.	100% of findings include a <code>revenue_at_risk_usd</code> value derived from reconciled metrics.	Must
FR-E2	Revenue-at-risk MUST be expressed as a counterfactual: dollars recoverable if the degraded rate returned to its <code>t0</code> / forecast baseline.	Actionable, comparable sizing.	Computation = $\Delta \text{rate} \times \text{affected_sessions} \times \text{downstream_value}$; method is documented and reproducible.	Must
FR-E3	Quantification SHOULD attach a confidence interval to the dollar estimate using the significance test inputs.	Honest uncertainty.	Findings include a low/high band on <code>revenue_at_risk_usd</code> .	Should

6.F Prescription / Experimentation

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-F1	The Prescribe agent MUST produce a prioritized experiment backlog, each item from <code>experiment-mcp.design_experiment(hypothesis, metric)</code> .	Statistically-defensible action.	Backlog items are test cards with hypothesis, target metric, and design; ordered by revenue-at-risk × confidence.	Must
FR-F2	Each experiment card MUST include a sample-size from <code>experiment-</code>	Feasibility and rigor.	Cards show required <code>n</code> , α , power, MDE, and	Must

ID	Requirement	Rationale	Acceptance Criteria	Priority
	<code>mcp.power_analysis(baseline, mde, alpha, power)</code> and a runtime from <code>runtime_estimate(n, traffic)</code> .		estimated days-to-significance from actual traffic.	
FR-F3	Prescriptions MUST tie each experiment to the specific finding and revenue-at-risk it addresses.	Traceability from diagnosis to action.	Every card references a finding ID and its <code>revenue_at_risk_usd</code> .	Must
FR-F4	The system SHOULD flag experiments that are underpowered given available traffic within the dataset window.	Avoid recommending infeasible tests.	Cards where <code>runtime_estimate</code> exceeds the window are marked "insufficient traffic".	Should

6.G Reporting / Decision Brief

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-G1	The Narrator agent MUST produce an executive "Decision Brief" via <code>report-mcp.render_brief(findings)</code> .	Proactive, executive-facing output.	Each run yields one brief covering top findings, each with cause, dollar impact, significance, and recommended action.	Must
FR-G2	The brief MUST be generated autonomously per scheduled run, not on demand only.	Anti-product: not an ad-hoc chatbot.	A brief is rendered and persisted on every scheduled run without human prompting.	Must
FR-G3	The system MUST persist briefs via <code>report-mcp.save_diagnosis(...)</code> and support <code>export(format)</code> .	Audit and distribution.	Briefs persist with run ID and timestamp; export supports at least markdown/JSON.	Must
FR-G4	The brief SHOULD surface prior-run context via <code>report-mcp.recall_prior(metric, segment)</code> to show trend continuity.	Learning narrative.	Recurring issues reference the prior diagnosis and whether the prescribed action shipped.	Should

6.H Memory & Learning

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-H1	The Memory store MUST persist prior diagnoses, a suppression list, the business glossary, and action-tracking.	Continuity and reduced re-work.	All four stores are queryable; each diagnosis is retrievable by (metric, segment, date).	Must
FR-H2	Action-tracking MUST record whether a prescribed experiment was implemented and its observed outcome.	Closed-loop learning.	Each prescription has a status $\in$ {open, shipped, won, lost, abandoned} with linkage to the post-test metric.	Should
FR-H3	The business glossary SHOULD map informal stakeholder terms to canonical metric/dimension names.	Robust conversational drill-down.	A glossary lookup resolves synonyms to canonical names; misses are logged for curation.	Could

6.I Evaluation

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-I1	An offline evaluation harness MUST grade root-cause diagnosis accuracy against a labeled benchmark.	Objective quality bar.	Harness reports root-cause accuracy; target $\geq 85\%$ vs $\leq 45\%$ naive baseline.	Must
FR-I2	The harness MUST assert 0 hallucinated columns/metrics (100% governed SQL) per run.	Grounding guarantee.	Any non-governed column/metric in executed SQL fails the eval.	Must
FR-I3	The harness MUST verify 100% of findings carry a significance test AND a dollar revenue-at-risk.	Enforce core principle.	Eval fails if any finding lacks either attribute.	Must
FR-I4	The eval harness MUST run in CI (GitHub Actions) alongside <code>dbt build</code> + tests.	Continuous correctness.	CI is red if accuracy/grounding/coverage thresholds are not met.	Must

6.J Orchestration & Scheduling

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-J1	The Orchestrator MUST run the plan-execute-critique loop across the seven agents in order: Orchestrator → Monitor → Decompose → Diagnose → Prescribe → Narrator, with Critic gating every finding.	Multi-agent control flow.	A run produces a trace showing each agent's invocation and the Critic verdict per finding.	Must
FR-J2	The system MUST run autonomously on a schedule (Cloud Scheduler / cron).	Autonomous operation, not on-demand.	A scheduled trigger executes a full run end-to-end with no human input.	Must
FR-J3	The Orchestrator SHOULD use Opus for orchestrate/critique and Sonnet for high-volume sub-tasks.	Cost-aware model routing.	Model assignment matches the policy and is recorded per agent call.	Should
FR-J4	A run MUST fail closed: if any guardrail (reconcile, dry-run cost, Critic) fails, affected findings are withheld, not shipped.	Trust.	Withheld findings appear in the trace with the failed gate, not in the brief.	Must

6.K Conversational Drill-Down (Secondary)

ID	Requirement	Rationale	Acceptance Criteria	Priority
FR-K1	The system MAY answer follow-up questions about a shipped brief, still routing all SQL through <code>semantic-mcp</code> and all math through <code>stats-mcp</code> .	Drill-down is secondary, not an ad-hoc SQL chatbot.	Conversational answers cite governed metrics only; no raw SQL is authored by the LLM.	Should
FR-K2	Drill-down MUST refuse questions requiring non-governed metrics/dimensions and explain the limitation.	Bounded, grounded interface.	Out-of-vocabulary requests return a refusal referencing <code>list_dimensions()</code> / <code>get_metric</code> .	Should
FR-K3	Drill-down COULD support natural-	Convenience.	"Why did this happen last week?" resolves to a stored diagnosis.	Could

ID	Requirement	Rationale	Acceptance Criteria	Priority
	language reference to prior diagnoses via Memory.			

7. Non-Functional Requirements

Non-functional requirements (NFRs) define the quality attributes Helios must satisfy. Each carries an NFR-ID, a measurable target consistent with the Foundation success metrics, and a verification method.

7.1 Performance / Latency

ID	Target	Verification
NFR-P1	Time-to-diagnosis < 5 minutes per autonomous run (end-to-end: Monitor → Narrator).	Timed CI run on the full window; p95 across runs < 5 min.
NFR-P2	<code>warehouse-mcp.dry_run(sql)</code> returns cost/bytes in < 2 s; <code>run_query</code> for any single governed query returns in < 30 s on the sample dataset.	Latency assertions in integration tests.
NFR-P3	Decomposition for a single dimension completes in < 1 s in <code>stats-mcp</code> .	Benchmark unit test on golden data.

7.2 Cost (BigQuery Byte & LLM Token Budgets)

ID	Target	Verification
NFR-C1	Total BigQuery bytes scanned per run MUST stay under a fixed byte budget; every query is gated by <code>dry_run</code> before execution.	Run aborts a query whose <code>dry_run</code> bytes exceed the per-query cap; run total is logged and asserted in CI.
NFR-C2	Incremental dbt + date-shard pruning MUST avoid full-table scans of <code>events_*</code> on routine runs.	Query plan shows only required <code>events_YYYYMMDD</code> shards scanned.
NFR-C3	LLM token spend per run MUST stay within a configured budget, using Opus only for	Per-agent token usage logged; run total asserted against budget.

ID	Target	Verification
	orchestrate/critique and Sonnet for high-volume sub-tasks.	

7.3 Reliability / Availability

ID	Target	Verification
NFR-R1	A scheduled run MUST complete or fail closed with a recorded reason; no silent partial briefs.	Trace shows terminal status; partial outputs never reach <code>save_diagnosis</code> as "complete".
NFR-R2	MCP tool calls MUST retry transient failures with bounded backoff (≤ 3 attempts).	Fault-injection test confirms retry then graceful failure.
NFR-R3	Re-running a failed run MUST be idempotent (no duplicate diagnoses for the same window).	Duplicate-run test asserts a single persisted diagnosis per (run window, finding).

7.4 Correctness & Trust Guardrails

ID	Target	Verification
NFR-T1	0 hallucinated columns/metrics — 100% of executed SQL is governed via <code>semantic-mcp</code> .	Eval harness (FR-I2) + static scan for raw <code>SELECT</code> .
NFR-T2	100% of shipped findings carry a significance test AND a dollar revenue-at-risk AND a recommended action.	Eval harness (FR-I3) gate.
NFR-T3	Every metric value reconciles to <code>warehouse-mcp.reconcile</code> within 0.5%.	Reconciliation assertion per finding.
NFR-T4	Every finding survives the Critic's refutation attempts before shipping.	Critic verdict recorded; "not survived" → withheld.

7.5 Security & PII / Privacy

ID	Target	Verification
NFR-S1	No PII egress. The dataset is obfuscated; <code>user_pseudo_id</code> is a device/cookie key only and MUST NOT be exported in briefs.	Output scan asserts no <code>user_pseudo_id</code> , raw IDs, or geo below region in exports.

ID	Target	Verification
NFR-S2	BigQuery access MUST use least-privilege service-account credentials scoped to the public dataset and the project's transformed tables.	IAM review; no broad project-owner roles.
NFR-S3	Secrets (service-account keys, LLM API keys) MUST NOT appear in logs, traces, or the repo.	Secret-scanning in CI; redaction in the logging layer.

7.6 Observability (Logging, Tracing, Audit Trail)

ID	Target	Verification
NFR-O1	Every run MUST emit a full audit trail: each agent call, each MCP tool call with inputs/outputs, every executed SQL, and every Critic verdict.	Trace artifact persisted per run; replayable.
NFR-O2	Each finding MUST be traceable from Decision Brief → diagnosis → significance test → governed SQL → reconciled total.	Lineage walk-through on a sample finding in CI.
NFR-O3	Structured logs MUST include run ID, agent name, model used, tokens, bytes scanned, and latency.	Log schema validated.

7.7 Reproducibility / Determinism

ID	Target	Verification
NFR-D1	All statistical computation is deterministic and lives in <code>stats-mcp</code> code, never in the LLM; identical inputs yield identical outputs.	Repeated decomposition/significance calls produce bit-stable results.
NFR-D2	A pinned run (fixed dataset window, seeds, model versions) MUST reproduce the same findings set.	Two pinned runs compared; finding set is identical.
NFR-D3	dbt models MUST be deterministic given the same source shards.	<code>dbt build</code> re-run yields identical fact rows.

7.8 Maintainability

ID	Target	Verification
NFR-M1	Adding a new governed metric MUST require only a semantic-layer definition + dbt model, no agent code change.	New-metric exercise touches only <code>models/semantic</code> and the registry.
NFR-M2	All models, metrics, and dimensions MUST follow <code>snake_case</code> and the canonical naming.	Linters enforces names; CI fails on deviation.
NFR-M3	MCP server tool signatures MUST be the exact canonical names/tools and stable across releases.	Contract tests against the five servers' tool schemas.

7.9 Scalability

ID	Target	Verification
NFR-X1	The pipeline MUST scale to additional date shards (extending beyond the ~2020-11 to 2021-01 window) without query-cost blowup, via incremental + partition pruning.	Synthetic shard extension stays within byte budget per run.
NFR-X2	The architecture SHOULD support additional dimensions/metrics without rewriting the decomposition core.	New dimension added to <code>list_dimensions()</code> flows through <code>decompose_change</code> unchanged.

7.10 Data-Quality SLAs

ID	Target	Verification
NFR-Q1	dbt tests MUST enforce: unique <code>transaction_id</code> in <code>fct_orders</code> ; non-null session key (<code>user_pseudo_id</code> , <code>ga_session_id</code>); <code>session_conversion_rate</code> $\in [0,1]$; funnel monotonicity (<code>purchasing_sessions</code> $\leq$ <code>begin_checkout_sessions</code> $\leq$ <code>add_to_cart_sessions</code> $\leq$ <code>view_item_sessions</code> $\leq$ <code>sessions</code>).	dbt test suite green in CI.
NFR-Q2	Rows with null <code>ga_session_id</code> or malformed <code>event_params</code> MUST be quarantined and counted, not silently dropped.	Quarantine table row counts reported; drop-rate threshold alarmed.
NFR-Q3	<code>revenue</code> (sum <code>purchase_revenue_in_usd</code>) and <code>transactions</code> (distinct <code>transaction_id</code>) MUST reconcile to canonical totals.	<code>reconcile(metric, grain)</code> parity check.

7.11 Portability

ID	Target	Verification
NFR-Z1	Python MCP servers MUST run on Python 3.11 with pinned dependencies (scipy, statsmodels, prophet/pmdarima, pandas, google-cloud-bigquery).	Lockfile build reproduces the environment in CI.
NFR-Z2	The system MUST run both in GitHub Actions CI and via the scheduler (Cloud Scheduler/cron) using the same containerized entrypoint.	One image runs in both contexts; no environment-specific code path.
NFR-Z3	Warehouse access SHOULD be abstracted behind <code>warehouse-mcp</code> so the BigQuery backend can be swapped without changing agent logic.	Agents reference only MCP tools, never the BigQuery client directly.